- Ass 3 due Fri 9 Oct @ 4pm

Recap:

(1) For testing a composite null hypothesis $H_0: \underline{\theta} \in \Theta_0$ versus a composite alternative $H_1: \underline{\theta} \in \Theta_1$ (where Θ_0 & Θ_1 are disjoint), the generalized likelihood ratio test (GLRT) statistic is

$$\Lambda = \frac{\sup_{\underline{\theta} \in \Theta_0} L(\underline{\theta} | \underline{X})}{\sup_{\underline{\theta} \in \Theta_1} L(\underline{\theta} | \underline{X})}$$

← best possible model under H_0

← best possible model under

(2) The GLRT rejects H_0 in favour of H_1 for "sufficiently small" values of Λ

Example 7.3 (comparing two group means, continued)

Consider two indep samples:

$$\underline{X}_1 = (X_{11}, X_{12}, \ldots, X_{1n_1})^T \overset{iid}{\sim} N(\mu_1, \sigma^2)$$

$$\underline{X}_2 = (X_{21}, X_{22}, \ldots, X_{2n_2})^T \overset{iid}{\sim} N(\mu_2, \sigma^2)$$

Suppose we are testing $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq \mu_2$

Numerator of GLRT: Under $H_0: \mu_1 = \mu_2$, recall that the MLE of the common mean is $\hat{\mu}_1 = \hat{\mu}_2 = \bar{X}_{..}$, the grand sample mean. And so the maximal likelihood achieved is

$$\sup_{H_0: \mu_1 = \mu_2} L(\mu_1, \mu_2 | \underline{X}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_1+n_2} \exp\left\{-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n_1}(X_{1i} - \bar{X}_{..})^2 + \sum_{i=1}^{n_2}(X_{2i} - \bar{X}_{..})^2\right]\right\}$$

In the exponent, we have

$$\underbrace{\sum_{i=1}^{n_1}(X_{1i}-\bar{X}_{..})^2}_{①} + \underbrace{\sum_{i=1}^{n_2}(X_{2i}-\bar{X}_{..})^2}_{②} \quad \left[\text{Total sum-of-squares}\right]$$

$$① = \sum_{i=1}^{n_1}\left(\underbrace{X_{1i}-\bar{X}_{1.}}_{a} + \underbrace{\bar{X}_{1.}-\bar{X}_{..}}_{b}\right)^2$$

$$= \text{from last slide}$$

$$= \sum_{i=1}^{n_1}\underbrace{(X_{1i}-\bar{X}_{1.})^2}_{a^2} + n_1\underbrace{(\bar{X}_{1.}-\bar{X}_{..})^2}_{b^2} \qquad 2ab = 0 \text{ exactly.}$$

Similarly, $② = \sum_{i=1}^{n_2}(X_{2i}-\bar{X}_{2.})^2 + n_2(\bar{X}_{2.}-\bar{X}_{..})^2$

So exponent = Total SS = $\overbrace{\sum_{i=1}^{n_1}(X_{1i}-\bar{X}_{1.})^2 + \sum_{i=1}^{n_2}(X_{2i}-\bar{X}_{2.})^2}^{\text{within group sum of squares}}$

$$\underbrace{+\, n_1(\bar{X}_{1.}-\bar{X}_{..})^2 + n_2(\bar{X}_{2.}-\bar{X}_{..})^2}_{\text{between group sum of squares.}}$$

Denominator of GLRT: Under $H_1: \mu_1 \neq \mu_2$, the MLEs for μ_1, μ_2 are:

$$\hat{\mu}_1 = \bar{X}_{1.} \qquad \hat{\mu}_2 = \bar{X}_{2.}$$

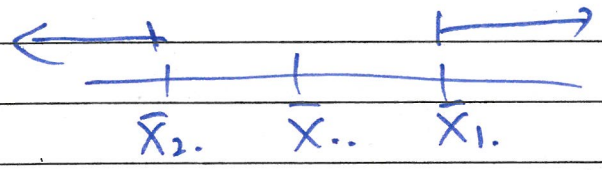
And so

$$\sup_{\mu_1 \neq \mu_2} L(\mu_1, \mu_2 \mid \underline{X}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{n_1+n_2} \exp\left\{-\frac{1}{2\sigma^2}\left[\underbrace{\sum_{i=1}^{n_1}(X_{1i}-\bar{X}_{1.})^2 + \sum_{i=1}^{n_2}(X_{2i}-\bar{X}_{2.})^2}_{\text{within group SS}}\right]\right\}$$

So the GLRT simplifies to

$$\Lambda = \frac{\text{numerator}}{\text{denominator}} = \exp\left[-\tfrac{1}{2\sigma^2}\left[n_1(\bar{X}_{1\cdot}-\bar{X}_{\cdot\cdot})^2 + n_2(\bar{X}_{2\cdot}-\bar{X}_{\cdot\cdot})^2\right]\right]$$

This ratio Λ is small when the sample means $\bar{X}_{1\cdot}$ & $\bar{X}_{2\cdot}$ are further apart



So the GLRT simplifies to the decision rule that rejects $H_0: \mu_1 = \mu_2$ in favor of $H_1: \mu_1 \neq \mu_2$ if

$$\left\{ \bar{X}_{1\cdot} - \bar{X}_{2\cdot} \leq d_1 \quad \text{OR} \quad \bar{X}_{1\cdot} - \bar{X}_{2\cdot} \geq d_2 \right\}$$

for some d_1 & d_2

How do we set critical values d_1 & d_2?

$$\bar{X}_{1\cdot} - \bar{X}_{2\cdot} \sim N\left(\mu_1 - \mu_2, \underbrace{\sigma^2/n_1 + \sigma^2/n_2}_{\sigma^2(1/n_1 + 1/n_2)}\right)$$

So under the null $H_0: \mu_1 - \mu_2 = \Delta_0$, we can set

$$d_1 = \Delta_0 - z^{(1-\alpha/2)}\,\sigma\sqrt{1/n_1 + 1/n_2} \quad \left(\begin{array}{c}2\text{-sample}\\ z\text{-test}\end{array}\right)$$

$$\& \quad d_2 = \Delta_0 + z^{(1-\alpha/2)}\,\sigma\sqrt{1/n_1 + 1/n_2}$$

In practice, we rarely know the pop^n σ^2. What would be our best estimate of σ^2?

$$\hat{\sigma}^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = S^2_{pooled}$$

weighted average of two sample variances

& we use t-quantiles instead of z-quantiles:

$$d_1 = \Delta_0 - t^{(1-\alpha/2)}_{n_1+n_2-2} S_{pooled}\sqrt{1/n_1 + 1/n_2}$$

& $$d_2 = \Delta_0 + t^{(1-\alpha/2)}_{n_1+n_2-2} S_{pooled}\sqrt{1/n_1 + 1/n_2}$$

We have formally constructed the famous <u>two-sample t-test</u>!

How do we extend this result to compare more than two groups at a time? We will look at ANOVA in week 11 (§10 of reference notes).

§ 8 Goodness-of-fit tests (for categorical data)

Recall from 1st year, when we have categorical data (e.g. contingency table of counts, where we cross-classify each unit according to levels of one or more factors), we used a goodness-of-fit test of the form:

$$\sum_{\text{cells } i} \frac{(\text{observed}_i - \text{expected}_i)^2}{\text{expected}_i} \overset{\text{approx}}{\sim} \chi^2_{\text{dist.}}$$

Where does this test come from?

Suppose each experiment unit can be classified as one of K categories (ie. one of K levels of a factor). For example,

handedness = {L, R, ambidextrous}

income brackets = {0-6K, 6K-15K, 15K-30K, ..., 150+K}

Popn: The underlying popn proportn of each level of a factor is denoted by

$$p_1, p_2, p_3, \ldots, p_K$$

where of course

$$p_K = 1 - p_1 - p_2 - p_3 \ldots - p_{K-1}$$

& so there are $K-1$ free parameters in general

Sample: Now suppose $X_1, X_2 \ldots, X_K$ are the counts in each category for a sample of n units.

Of course, $X_1 + X_2 + \ldots + X_K = n$

& so $X_K = n - X_1 - X_2 - \ldots - X_{K-1}$

Model?

The pmf of $\underset{\sim}{X} = (X_1, X_2 \ldots, X_K)'$ ~ multinomial $(p_1, p_2 \ldots, p_K)$ is

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_K = x_K)$$

$$= \binom{n}{\underset{\sim}{x}} p_1^{x_1} p_2^{x_2} p_3^{x_3} \ldots p_K^{x_K}$$

where $\binom{n}{\underset{\sim}{x}} = \dfrac{n!}{x_1!\, x_2! \ldots x_K!}$ is the multinomial coefficient.

As a special case, $K = 2$ categories then

$$P(X_1 = x_1, X_2 = x_2) = \frac{n!}{x_1!\, x_2!} p_1^{x_1} p_2^{x_2}$$

$$= \frac{n!}{x_1!\,(n-x_1)!} p_1^{x} (1-p_1)^{n-x}$$

~ binomial pmf ✓

Suppose for now we are interested in testing a single point null hypothesis:

$$H_0: \underset{\sim}{p} = (p_1, p_2, \ldots, p_K) = \underset{\sim}{p}^* = (p_1^*, p_2^*, \ldots, p_K^*)$$

where $\underset{\sim}{p}^*$ is a given set of proportions

(e.g. $\underset{\sim}{p}^* = (0.10, 0.85, 0.05)$ for handedness)

versus $H_1: \underset{\sim}{p} \neq \underset{\sim}{p}^*$

NOTE: The alternative is NOT the same as $p_1 \neq p_1^*, p_2 \neq p_2^*, \ldots, p_K \neq p_K^*$

but rather, at least one proportion $p_K \neq p_K^*$ for some k

Consider the GLRT statistic:

$$\Lambda = \frac{\sup_{H_0} L(\underset{\sim}{p} | \underset{\sim}{X})}{\sup_{H_1} L(\underset{\sim}{p} | \underset{\sim}{X})}$$

numerator: $\sup_{H_0} L(\underset{\sim}{p} | \underset{\sim}{X}) = L(\underset{\sim}{p}^* | \underset{\sim}{X})$

$$= \binom{n}{\underset{\sim}{x}} (p_1^*)^{x_1} (p_2^*)^{x_2} \cdots (p_K^*)^{x_K}$$

denominator: what would be the MLEs for each $p_1, p_2, \ldots, p_K$ under the alternative hypothesis H_1?

$$\hat{p}_1 = \frac{x_1}{n}, \quad \hat{p}_2 = \frac{x_2}{n}, \quad \ldots, \quad \hat{p}_K = \frac{x_K}{n},$$

precisely the sample proportion of each category

So $\sup_{H_1} L(p|x) = \binom{n}{x_1}\left(\hat{p}_1\right)^{x_1}\left(\hat{p}_2\right)^{x_2}\cdots\left(\hat{p}_k\right)^{x_k}$

$= \binom{n}{x}\left(\frac{x_1}{n}\right)^{x_1}\left(\frac{x_2}{n}\right)^{x_2}\cdots\left(\frac{x_k}{n}\right)^{x_k}$

$\Rightarrow$ so $\Lambda = \left(\frac{p_1^*}{\hat{p}_1}\right)^{x_1}\left(\frac{p_2^*}{\hat{p}_2}\right)^{x_2}\cdots\left(\frac{p_k^*}{\hat{p}_k}\right)^{x_k}$

small values of $\Lambda \Leftrightarrow$ small values of $\log\Lambda \Leftrightarrow$ large values of $-\log\Lambda$

where $-\log\Lambda = x_1\log\left(\frac{\hat{p}_1}{p_1^*}\right) + x_2\log\left(\frac{\hat{p}_2}{p_2^*}\right) + \cdots + x_k\log\left(\frac{\hat{p}_k}{p_k^*}\right)$

$= x_1\log\left(\frac{x_1}{np_1^*}\right) + x_2\log\left(\frac{x_2}{np_2^*}\right) + \cdots x_k\log\left(\frac{x_k}{np_k^*}\right)$

$= O_1\log\left(\frac{O_1}{E_1}\right) + O_2\log\left(\frac{O_2}{E_2}\right) + \cdots + O_k\log\left(\frac{O_k}{E_k}\right)$

where O_k's are observed counts from category k
& E_k's are expected counts from category k.

$= \sum_{\text{cells } k} O_k\log\left(\frac{O_k}{E_k}\right)$

This is the exact GLRT for categorical data!

BUT, why do we teach

Pearson's $\chi^2 = \sum_{\text{cells } k}\frac{(O_k - E_k)^2}{E_k}$ instead of

$\text{GLRT} = \sum_{\text{cells}}^{K} O_k\log\left(\frac{O_k}{E_k}\right)$ in first year?

For (historical) reasons:

① The null dist of the exact GLRT $-\log\Lambda$ is not easily derived, (& so cutoffs are hard to derive)

② Twice the GLRT $-2\log\Lambda = 2\sum_{\text{cells}} O_k \log\left(\frac{O_k}{E_k}\right)$

is asymptotically equivalent to Pearson's $\chi^2 = \sum_{\text{cells}} \frac{(O_k - E_k)^2}{E_k}$

In fact, Pearson's χ^2 is actually the first-order term in a Taylor expansion of $-2\log\Lambda$

③ The null dist of Pearson's χ^2 is relatively easy to derive as Pearson's $\chi^2 \longrightarrow \chi^2$ dist

(Note ② + ③ $\Rightarrow$ $-2\log\Lambda \overset{\text{approx}}{\sim} \chi^2_{\text{dist}}$)

④ In 1900, Pearson's χ^2 is far easier to compute without a calculator.

Example 8.4 (Mendel's genetic data). In famous paper, Mendel (1865) crossed yellow, smooth male peas with green, wrinkly female peas. According to Mendel's genetics theory, the relative frequency of the progeny should be:

Type	Mendel's Theory:	Mendel's counts
smooth yellow	9/16	315
smooth green	3/16	108
wrinkly yellow	3/16	102
wrinkly green	1/16	31
	16/16 ✓	556 ✓

To test this theory, Mendel collected 556 progenies & classified them accordingly. The counts are given in green above

Are Mendel's counts consistent with the relative frequencies predicted by his theory?

Here, $H_0: p = (9/16, 3/16, 3/16, 1/16) = p^*$

versus $H_1: p \neq p^*$, or at least one of the proportions is wrong

If H_0 is true, then what counts would we have expected in each category?

$E_1 = n p_1^* = 556 \times 9/16 = 312.75$

$E_2 = 556 \times 3/16 = 104.25$

$E_3 = 556 \times 3/16 = 104.25$

$E_4 = 556 \times 1/16 = 34.75$

556.00

The exact GLRT statistic here is

$$-2\log\Lambda = 2\sum_{\text{cells}} O_k \log\left(\frac{O_k}{E_k}\right)$$

$$= 2\left[315\log\left(\frac{315}{312.75}\right) + 108\log\left(\frac{108}{104.25}\right) + 102\log\left(\frac{102}{104.25}\right) + 31\log\left(\frac{31}{34.75}\right)\right]$$

$$= 0.618$$

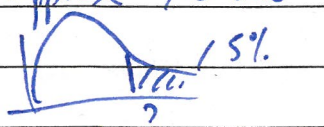
Pearson's χ^2

$$= \sum_{\text{cells}} \frac{(O_k - E_k)^2}{E_k}$$

$$= \frac{(315-312.75)^2}{312.75} + \frac{(108-104.25)^2}{104.25} + \frac{(102-104.25)^2}{104.25} + \frac{(31-34.75)^2}{34.75}$$

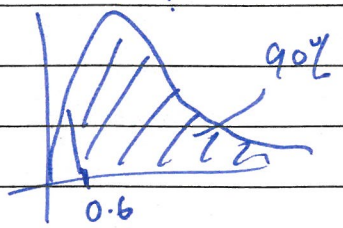
$$= 0.604$$

Under H_0, these statistics both have an approx χ^2 dist with $df = K-1 = 3$



So p-value for the test is

$$P(\chi^2_3 \geq 0.604\ (0.618)) \simeq 90\%.$$



Conclusion (for now?) Mendel's counts are consistent with his theory.
(p-value = 0.9) ✓

Can we say that the data SUPPORTS Mendel's Theory?

NO! Who knows, suppose Alan's refine Mendel's Theory to:

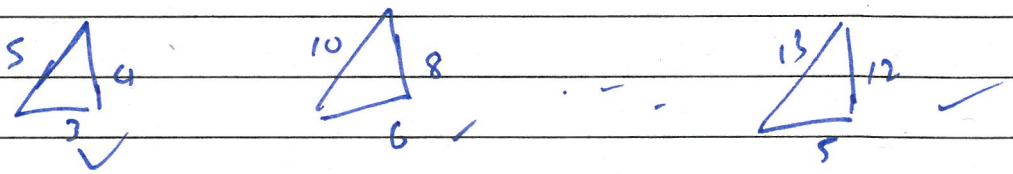
$$P_{Alan} = \left(9.01/16, \; 3.02/16, \; 2.98/16, \; 0.99/16 \right)$$

& the same dataset would now "support" Alan's theory as well.

Philosophy corner: Can we ever find evidence FOR a null hypothesis through random experiments?

Example 1: Pythagoras Theorem claims $H_0: c^2 = a^2 + b^2$ for all right-angle triangles.

We can sample as many right angle triangles as we want:



BUT we cannot support/prove Pythagoras's Theorem NO MATTER how many triangles we sample.

Example 2: In Victorian England, it was widely believed that all swans are white, & the more white swans people spotted the more they believed this claim to be true (!) But of course, we know black swans exist - Western Australia, & so a single counter-example disproves this claim